

Ridge Regression — Other Important Points

- **Used for:** Regression problems where overfitting or highly correlated features are present.
- **L2 Regularization:** Ridge uses the sum of squared coefficients as its penalty.
- **Multicollinearity:** Ridge works well when input features are highly correlated.
- **Hyperparameter:** alpha (or λ) controls the strength of regularization.
- **Standardization:** Features should usually be scaled before applying Ridge because the penalty depends on coefficient size.
- **Prediction:** After training, use `.predict()` just like Linear Regression.
- **Evaluation:** Common metrics are MAE, MSE, RMSE, and R^2 .
- **Scikit-learn:**

```
from sklearn.linear_model import Ridge
```

```
model = Ridge(alpha=1)
```

```
model.fit(X_train, y_train)
```

```
y_pred = model.predict(X_test)
```

Ridge Regression — Short Notes

1. **Coefficient Shrinkage:**
As λ increases, coefficients become smaller, but generally **not exactly 0**.
2. **Large coefficients:**
Large weights are penalized more because Ridge uses .
3. **Bias–Variance:**
 $\lambda \uparrow \rightarrow \text{Bias } \uparrow, \text{ Variance } \downarrow$.
Too much $\lambda \rightarrow$ underfitting; too little $\rightarrow$ overfitting.
4. **Loss Function:**
5. **Key idea:**
Ridge = L2 regularization $\rightarrow$ penalizes large coefficients $\rightarrow$ reduces overfitting.